AI-POWERED LOCAL AGRI-MARKET INTELLIGENCE AND FARMER-BUYER CONNECT SYSTEM

A Thesis
Presented to
The Academic Faculty

By

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In Partial Fulfillment
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ABSTRACT

Agriculture is a vital part of the Indian economy because it provides jobs for more than half of the country's workers and contributing significantly to food security. Despite advances in agricultural production, Indian farmers — particularly those at the small and medium scale — continue to face severe challenges in post-harvest market access, price discovery, and buyer connectivity. The absence of real-time, localized, and actionable market intelligence forces farmers to depend on intermediaries, resulting in below-market realizations and reduced profitability.

This thesis presents the design, architecture, and implementation of an AI-Powered Local Agri-Market Intelligence and Farmer-Buyer Connect System — a multi-agent, bilingual, explainable artificial intelligence platform that enables farmers to access real-time agricultural market intelligence through natural language queries in Hindi and English. The system addresses the critical gap between available government mandi data and the farmer's ability to use this information to make smart choices.

The proposed system integrates a ReAct (Reasoning and Acting) AI Agent powered by the Gemini 2.5 Flash Large Language Model with a Multi-Agent Architecture comprising ten specialized agents: Intent Detection, Location Resolution, Mandi Intelligence, Tavily Internet Search, Reasoning, Price Prediction, Fact-Check, Answer Generation, and Sell Decision agents. The system employs Retrieval-Augmented Generation (RAG) using a FAISS vector database with sentence-transformer embeddings to retrieve relevant historical price context, augmenting the LLM's reasoning with verified factual data.

Live mandi price data is fetched from the Government of India's Agmarknet API and eNAM platform, which serve as the trusted source of truth. The Tavily Search tool enables the agent to retrieve current internet-based market trends, agricultural news, and demand signals. A dedicated Fact-Check Agent cross-validates AI-generated

insights against government data and search results, assigning confidence scores to prevent hallucination. A Prediction Agent employs moving-average trend analysis to forecast price trends and generate sell-or-wait recommendations with quantified confidence.

The system's bilingual Answer Generation Agent produces farmer-friendly explanations in both Hindi and English, ensuring accessibility across linguistic demographics. The frontend, built with React.js and Tailwind CSS, features a voice-enabled AI chatbot interface, animated mandi dashboards, trend charts, and an Explainable AI reasoning panel. The backend system is organized and managed using Node.js with Express.js and MongoDB, while the AI microservice layer is implemented using Python and FastAPI.

Experimental results show that the system correctly understands what the user wants and finds the right local market prices, performs multi-market comparative reasoning, and quickly creates helpful replies in bilingual responses based on the user's specific situation. The architecture is modular, scalable, and production-ready, suitable for deployment on cloud infrastructure via Vercel, Render, and MongoDB Atlas.

This research contributes to the fields of Agentic AI, Agricultural Informatics, and Explainable AI by demonstrating that a multi-agent ReAct framework grounded in government-verified data can bridge the digital divide in agricultural market access and empower rural farmers with intelligent, trustworthy, and accessible decision support.

Keywords: Agentic AI, ReAct Agent, Multi-Agent System, RAG, FAISS, Mandi Intelligence, Agricultural Market, Gemini LLM, NLP, Explainable AI, Bilingual AI, Farmer-Buyer Connect.

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LIST OF SYMBOLS AND ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
APMC	Agricultural Produce Market Committee
CORS	Cross-Origin Resource Sharing
eNAM	Electronic National Agriculture Market
FAISS	Facebook AI Similarity Search
FastAPI	Fast Application Programming Interface (Python web framework)
GPS	Global Positioning System
IIT	Indian Institute of Technology
IP	Internet Protocol
JSON	JavaScript Object Notation
LLM	Large Language Model
MAS	Multi-Agent System
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
ReAct	Reasoning and Acting (AI agent framework)
REST	Representational State Transfer
UI	User Interface
XAI	Explainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

Agriculture is a foundational part of India's economy — it provides livelihoods for over 58 percent of the country's workers and accounts for nearly one-fifth of national GDP. With more than 140 million farming families, the sector carries both economic weight and deep social significance. But numbers alone don't capture the everyday reality of most of these farmers. The majority run small or medium-sized farms. They have limited access to credit, limited reach into broader markets, and — critically — limited access to the kind of timely, reliable information that could help them sell their crops at a fair price.

The problem isn't production. India's agricultural output has grown considerably over the decades, driven by improved seeds, irrigation, and mechanization. The gap, rather, is in what happens after the harvest. Farmers often have no way of knowing which nearby market is offering the best price for their crop that day, or whether prices are expected to rise or fall in the coming week. Without this information, most end up selling through local middlemen — often at prices well below what the market would actually support.

Although we have seen the roll out of many digital initiatives and introduction of government platforms which are supposed to close this gap and modernize agricultural trade, but to access and to use of these is still very much difficult. Government platforms such as Agmarknet and eNAM, which connects physical mandis into a unified digital trading platform and publish daily wholesale mandi prices across thousands of markets. These are genuine achievements in agricultural data infrastructure. But only issue is that the information is in technical terms which the farmers cannot understand in easy way this does not make sense of use. Agmarknet shows dense tabular records, with no personalization, no trend analysis, and no

guidance on what a farmer should actually do with the information. eNAM, despite its scale, has struggled with adoption among small farmers who find the interface confusing and the process unfamiliar.

Also, it is a known fact that the large majority of farmers prefer to communicate in their native languages which include Hindi, Bhojpuri, Bengali, Gujarati, Marathi, Telugu, etc. We see a large accessibility gap because the digital interfaces are not localized, language friendly and decision supportive which in turn leaves farmers out of the benefits of the available digital resources.

Mainly in the rural and semi-urban areas, we are seeing above problem due to low internet penetration, less awareness of digital platforms, and no access to analysis tools. This causes farmers to sell the crops to middlemen at below market price which reduces profit and which also rules out sustainable practices. We observed a great need for a system which transforms raw mandi data into easy-to-use real time information which is also custom for each farmer's location, local language and crop requirements.

To solve the above problem, this project an **AI-Powered Local Agri-Market Intelligence and Farmer-Buyer Connect System**, is designed to result as smart and context aware interface for farmers. The system will enable farmers to ask questions in natural or local language like:

- "गेहूं बेचने का सबसे अच्छा समय क्या है?"
- "मुझसे टमाटर कौन खरीद रहा है?"

And answer will be data-based responses that will be easily understandable by farmers and also information will be based on real-time market conditions.

The emergence of Large Language Models, agentic AI frameworks, and retrieval-augmented generation presents an unprecedented opportunity to bridge this gap. These technologies, when designed with farmer-centric principles — multilingual support, explainability, and grounding in verified government data — can transform raw mandi data into actionable, contextualized intelligence that farmers can trust and act upon.

This research is motivated by the observed reality that over 60 percent of Indian farmers in semi-urban and rural areas primarily communicate in regional languages such as Hindi, Bhojpuri, Awadhi, and Maithili, yet no existing platform provides a conversational AI interface that combines live government mandi data, multi-agent reasoning, and bilingual explainable responses in a single unified system.

1.2 Problem Statement

The primary problem addressed in this research is the inaccessibility of real-time, localized, and explainable agricultural market intelligence for small and medium-scale farmers in India. Specifically, the following deficiencies characterize the current landscape:

- Existing government platforms like Agmarknet and eNAM provide raw price tables with no interpretation, no trend analysis, and no recommendation on whether to sell or wait.
- No existing conversational interface supports farmers' market queries in Hindi or regional languages with real-time mandi data integration.
- Price information is not geographically personalized — farmers don't have visibility into which nearby mandi is offering the best price for their specific crop right now.
- The absence of AI-driven reasoning means farmers cannot obtain structured advice such as whether current prices represent a good selling opportunity relative to the recent trend.
- Hallucination risk in general-purpose LLMs makes them unsuitable for agricultural price queries without a grounded, fact-checking architecture.
- Farmers have to check multiple websites or apps just to do business. Information about buyers and market contact details are kept completely separate from crop prices, instead of being available in one single place.

1.3 Research Objectives

The specific objectives of this research are as follows:

- To design and implement a multi-agent AI architecture capable of orchestrating specialized agents for intent detection, location resolution, mandi price retrieval, trend prediction, fact verification, and bilingual answer generation.
- To integrate live government mandi price data from Agmarknet and eNAM as the primary source of truth, preventing the LLM from fabricating price information.
- To implement a ReAct (Reasoning and Acting) AI agent powered by Gemini 2.5 Flash LLM using LangChain's AgentExecutor framework with dynamic tool selection.
- To build a FAISS-based vector database for RAG that retrieves relevant historical mandi context to augment LLM reasoning.
- To develop a Tavily-integrated internet search capability that enables the agent to access current market trends, demand signals, and agricultural news.
- To implement a bilingual answer generation module producing farmer-friendly explanations in Hindi and English with transparent reasoning steps.
- To design a React.js frontend with voice input/output, animated mandi dashboards, trend charts, and an Explainable AI panel.

1.4 Scope of the Work

This research focuses on the design, implementation, and evaluation of an AI-powered agricultural market intelligence system targeting farmers in Bihar, Uttar Pradesh, Madhya Pradesh, Punjab — the broader Hindi-speaking belt of India. The system handles four primary query intents: price query, buyer search, sell advice, and trend analysis. Crop coverage spans major commodities traded in India mandis, including tomato, onion, wheat, potato, rice, lentils etc.

The system integrates data from the Government of India's data.gov.in API (Agmarknet dataset) and eNAM platform for market and buyer information. Location detection supports GPS, IP-based geolocation, pincode extraction, and state/district extraction from natural language queries. The deployment target encompasses cloud infrastructure with Vercel for the frontend, Render for backend and AI services, and MongoDB Atlas for data persistence.

1.5 Organization of the Thesis

The remainder of this thesis is organized as follows. Chapter 2 presents a comprehensive review of existing literature spanning digital agriculture platforms, multi-agent systems, NLP for agriculture, and AI-driven price forecasting, closing with a structured analysis of research gaps. Chapter 3 describes the overall system design and three-tier architecture covering the frontend, backend, and AI service layers. Chapter 4 details the methodology and implementation of each system component, from the NLP pipeline and ReAct agent to the RAG pipeline and bilingual answer generation. Chapter 5 presents the results of system evaluation across performance, accuracy, and qualitative dimensions. Chapter 6 concludes the thesis with a summary of contributions, identified limitations, and directions for future research.

CHAPTER 2

LITERATURE REVIEW

This chapter surveys the existing body of research relevant to the proposed system across six thematic areas: digital agriculture and market information systems, government agricultural platforms, farmer-buyer digital marketplaces, multi-agent systems in agriculture, NLP and conversational AI for agriculture, and AI-driven price prediction and decision support. The chapter concludes with a structured gap analysis that motivates the research contributions of this thesis.

2.1 Digital Agriculture and Market Information Systems

The intersection of information and communication technologies with agricultural market systems has been a significant area of research since at least the early 2000s. Fafchamps and Minten demonstrated in 2012 showed that access to price information via SMS meaningfully improved the bargaining power of Indian farmers and reduced price dispersion across markets. Their research proved that when farmers have access to market information, their income goes up. This discovery gave creators a strong reason to build digital tools and apps that share market data with farmers.

At the national level, India's Agmarknet platform, launched by the Directorate of Marketing and Inspection under the Ministry of Agriculture and Farmers Welfare, aggregates daily wholesale mandi prices from over 3,500 markets across the country. While this represents a landmark in agricultural data digitization. But research by Kaur and Kaur (2020) identified critical limitations: the platform lacks personalization, trend analysis, predictive capability, and conversational accessibility. Farmers visiting Agmarknet encounter dense tabular data that offers no personalization, no trend analysis, no predictive capability, and no way to ask a question. Farmers land on a page of tables and have to figure out what to do with them.

The Food and Agriculture Organization's Agricultural Market Information System (AMIS), established in 2011, similarly aggregates global commodity price data

for policy-level decision making. However, as Katchova and Enlow (2013) noted, such macro-level systems provide little utility to individual farmers making crop-specific, location-sensitive selling decisions. The literature consistently identifies a gap between aggregated market data and the farmer's need for localized, actionable, and conversational intelligence.

2.2 Government Agricultural Platforms

The Government of India has launched several digital initiatives aimed at modernizing agricultural trade. The electronic National Agriculture Market (eNAM), launched in 2016, connects physical mandis into a unified online trading platform and has enrolled over 1.77 crore farmers across 1,260 mandis as of 2023. Chand (2017) analysed eNAM's early adoption challenges, identifying that small farmers often avoided using the platform. This happened because they did not get enough help learning how to use the digital tools, found the app screens too complicated, and lacked experts to guide them through the process.

Other government initiatives including Kisan Suvidha, IFFCO Kisan, and mKisan provide agricultural advisory via SMS and mobile notifications. However, these systems deliver static price lists without dynamic reasoning, prediction, or conversational support. Research by Tripathi, Mishra, and Shukla (2022) highlighted that the next generation of agricultural tools must embed intelligence within information delivery itself — the shift from showing data to interpreting it.

A theme that runs through virtually all this literature is the language accessibility problem. Most digital interfaces operate exclusively in English or formal Hindi, whereas the majority of target farmers communicate in regional dialects such as Bhojpuri, Awadhi, Maithili, Marathi, Telugu, and dozens of other regional varieties. This linguistic barrier effectively excludes a large segment of the farmer population from benefiting from available digital resources, and it’s forcing the need for multilingual conversational interfaces.

2.3 Farmer-Buyer Digital Marketplaces

Private digital marketplaces have emerged as an alternative to government platforms in connecting farmers with buyers. Platforms such as ITC's e-Choupal, Ninjacart, DeHaat, and BigHaat have demonstrated measurable improvements in farmer income through technology-enabled supply chain integration. Jain and Kumar (2021) reviewed AI applications in Indian agriculture and found that farmers using digital procurement platforms gained better bargaining power, reduced post-harvest losses, and faster sales cycles.

However, a common criticism of private marketplace platforms is that they function primarily as procurement tools serving corporate buyers rather than as open decision-support systems empowering farmers. Price transparency is often limited — farmers may receive offers without visibility into alternative market options, comparative price benchmarks, or trend-based advice on optimal selling timing. The World Bank's 2020 report on digital agriculture in India made a specific call for unbiased, AI-based advisory systems that genuinely empower farmers' independent decision-making, rather than systems that channel farmers toward particular buyers.

2.4 Multi-Agent Systems in Agriculture

Multi-Agent Systems (MAS) have been applied to agricultural problems across a range of domains including precision farming, irrigation scheduling, crop disease identification, and supply chain optimization. Ferrag and Maglaras (2021) provided a comprehensive survey of deep learning approaches for agriculture and noted that agentic and multi-agent architectures are particularly well-suited to problems where information changes rapidly, data comes from multiple sources, and different decisions depend on each other — all of which describe agricultural market intelligence.

In the context of market news and data, these agent-based smart digital systems can do many jobs at once. These systems having capability to simultaneously monitor mandi data feeds, track price trends, They can watch live market feeds, track where

prices are going, find opportunities to buy low in one market and sell high in another, and give instant advice to users. The transition from passive dashboards to active agentic AI frameworks represents a significant qualitative shift in system capability. Recent literature (Bhattacharya, Rawat, and Kumar, 2021) emphasized the shift from passive dashboards to active agentic AI: instead of presenting a dataset that requires manual interpretation, intelligent agents can monitor data feeds continuously, detect opportunities, and deliver proactive, contextualized insights.

The LangChain ReAct (Reasoning and Acting) framework, introduced by Yao et al. (2023), established a standard method for building AI agents that can think and act at the same time. This allows the AI to use step-by-step logic in natural language while automatically pulling in outside tools when it needs extra help or data. Each reasoning step is visible and auditable, which significantly reduces hallucination requiring factual grounding in question-answering tasks. The adoption of the ReAct framework is the direct theoretical basis for the agent architecture in this research.

2.5 Natural Language Processing and Conversational AI for Agriculture

The application of NLP and conversational AI to agricultural advisory represents an active and growing field of research. Systems including Kisan GPT, FarmSpeak, and various institutional research prototypes from ICAR and IIT laboratories have demonstrated that farmer-facing chatbots increase platform engagement, reduce information barriers, and improve the adoption rate of digital advisory services. Nandhini and Suresh (2022) evaluated a Hindi-language agricultural chatbot and found that native language support was the single most important determinant of adoption among rural users.

However, the literature identifies several systematic limitations of existing agricultural chatbots. First, most systems provide only general advisory content — crop disease management, weather information, farming tips — without integration of real-time market price data. Second, very few systems support regional Indian dialects, limiting their reach to formally educated or urban-adjacent farmers. Third, the majority of existing systems lack transparency in their reasoning processes, which research consistently identifies as a barrier to trust among farmer users. Explainable AI (XAI)

remains an underexplored dimension of agricultural AI, despite its recognized importance for adoption.

2.6 AI-Driven Price Prediction and Decision Support

Machine learning and deep learning approaches to agricultural price forecasting have been extensively studied. Classical time-series methods including ARIMA and its variants have been applied to mandi price prediction with moderate success, but their assumption of stationarity limits performance under the volatile, externally-driven conditions characteristic of Indian agricultural markets. Bhattacharya et al. (2021) applied deep learning networks including LSTM and GRU architectures to predict crop price fluctuations in Indian mandis, reporting improvements over baseline statistical models.

Shah and Patel (2021) reviewed the role of AI in precision agriculture and identified price intelligence as one of four critical AI application domains, alongside crop disease detection, yield prediction, and precision input application. Their analysis highlighted the importance of integrating multi-source data — mandi prices, weather forecasts, transport costs, and seasonal patterns — for robust price forecasting. The RAG architecture adopted in this research directly addresses this multi-source integration challenge by combining government API data, vector-stored historical context, and live internet search within a single reasoning pipeline.

The importance of Explainable AI in agricultural decision support is a recurring theme in recent literature. Recent research shows that farmers need to understand how AI makes decisions before they will trust it. When an AI system explains the clear logic behind its advice, farmers are much more likely to trust the system and follow its recommendations. This fact is the exact reason why the new system includes an Explainable AI panel and shows its step-by-step thinking openly and transparent reasoning steps implemented in the proposed system.

2.7 Research Gap Analysis

Based on the reviewed literature, the following research gaps are identified that motivate this thesis:

Gap Area	Existing Limitation	This Research Addresses
Localized Intelligence	Generic national-level price lists	GPS/IP/pincode-based mandi filtering
Multilingual Access	English/formal Hindi only	Hindi-English bilingual NLP and response
Conversational Interface	Static dashboards and tables	ReAct agent with voice-enabled chatbot
Explainability	Black-box recommendations	XAI panel with step-by-step reasoning
Hallucination Prevention	LLM-generated unverified prices	Government API as source of truth + fact-check agent
Sell Decision Support	No sell/wait advisory available	Prediction agent with confidence-scored recommendation
Integrated Buyer Connect	Siloed from price intelligence	eNAM buyer data integrated in unified system

Table 2.1: Comparative Analysis of Existing Agricultural Platforms and Research Gaps

CHAPTER 3

SYSTEM DESIGN AND ARCHITECTURE

This chapter presents the comprehensive design and architectural blueprint of the AI-Powered Local Agri-Market Intelligence and Farmer-Buyer Connect System. The system follows a three-tier, IEEE-compliant intelligent system architecture comprising a Presentation Layer (React.js frontend), a Business Logic Layer (Node.js/Express.js backend), and an AI Intelligence Layer (Python/FastAPI multi-agent service). Each tier is designed to be modular, independently scalable, and clearly separated in its responsibilities.

3.1 Overall System Architecture

The system architecture is organized as three interdependent tiers communicating via REST APIs. The frontend tier provides the farmer-facing user interface; the backend tier handles authentication, query management, data aggregation, and API gateway functions; and the AI service tier hosts the multi-agent intelligence pipeline. Figure 3.1 illustrates the complete three-tier architecture.



Figure 3.1: Three-Tier System Architecture of AI Agri-Market Intelligence System

The architectural design adheres to the following principles: (1) it only uses official government data for accurate crop prices; (2) the AI model is used only to think and explain things, never to guess or provide price information; (3) the system is split into separate parts so each tool can be built and tested on its own; (4) the technology can easily scale up to handle many users at the same time; and (5) the app layout is designed to be simple and easy for farmers to use, rather than being complicated and technical.

3.1.1 Technology Stack

Layer	Technology	Purpose
Frontend	React.js, Tailwind CSS, Framer Motion	AI chatbot interface, voice, charts
	Recharts, Axios, React Router	Data visualization and API communication
Backend	Node.js, Express.js	REST API gateway and authentication
	MongoDB, Mongoose	Persistent query and user data storage
AI Service	Python 3.11, FastAPI, Uvicorn	Multi-agent orchestration and inference
	LangChain 0.2.16, Gemini 2.5 Flash	ReAct agent and LLM reasoning
	FAISS, HuggingFace Embeddings	Vector store for RAG pipeline
	Tavily Search API	Real-time internet market intelligence
Data Sources	Agmarknet API (data.gov.in)	Live government mandi prices
	eNAM Platform	Buyer and APMC contact information

Table 3.1: Technology Stack Summary

3.2 Frontend Architecture

The frontend is built with React.js and Tailwind CSS, implementing a sleek, glass-like look with green gradients, giving it a high-tech yet agricultural feel. The interface is fully mobile-responsive and supports both dark and light modes. To handle information securely, the frontend talks directly to the backend system using standard web requests, keeping the user logged in safely during their session. The frontend communicates with the Node.js backend exclusively via Axios-based REST calls, with JWT tokens stored in context for session management.

Key frontend components include: (1) the AI ChatBot interface with streaming response animation, voice-to-text input via the Web Speech API, and text-to-speech output; (2) the Live Mandi Dashboard featuring animated cards showing real-time crop prices from nearby mandis; (3) the Trend Chart component built with Recharts displaying historical price curves with directional indicators; (4) the Explainable AI Reasoning Panel that renders the step-by-step thought process of the ReAct agent; and

(5) the Confidence Score Indicator providing visual feedback on the reliability of each response.

The frontend directory structure follows a modular pattern: components/ for reusable UI elements (Navbar, Footer, ChatBot, MandiCard, TrendChart, ConfidenceBar, ReasoningPanel); pages/ for route-level components (HomePage, ChatPage, Dashboard, PricePage, LoginPage, ProfilePage); services/ for Axios API wrappers; hooks/ for custom React hooks including useVoice and useLocation; and context/ for global state management via AuthContext and QueryContext.

3.3 Backend Architecture

The Node.js/Express.js backend serves as the system's API gateway, managing user authentication via JWT, routing farmer queries to the AI service, and and stores interaction history in MongoDB. The backend exposes REST endpoints under five route namespaces: /auth for registration and login; /queries for query submission and history retrieval; /users for profile management; /mandis for cached mandi data serving; and /prices for pre-processed price trend data.

The MongoDB schema stores two primary models: User (storing name, phone number, preferred language, location, and query history) and Query (storing the original query text, detected intent, extracted entities, AI response, confidence score, and timestamp). This persistence layer enables the system to maintain contextual memory across sessions and supports future analytics on query patterns and farmer decision outcomes.

3.4 AI Service Architecture

The AI service, implemented in Python with FastAPI, constitutes the intellectual core of the system. It hosts the multi-agent pipeline that processes each farmer query through a sequence of specialized reasoning steps. The service exposes a primary endpoint at POST /agent/query, which accepts the query text, optional location

coordinates, and session context, and returns the structured response object detailed in Section 4.7.

3.4.1 Ten Specialized AI Agents

Agent	Role and Responsibility
ReAct Orchestrator	LangChain AgentExecutor powered by Gemini 2.5 Flash; orchestrates tool selection and multi-step reasoning in Thought-Action-Observation cycles
Intent Detection Agent	Classifies query into one of four intents: price_query, buyer_search, sell_advice, trend_analysis using hybrid rule-based and keyword NLP
Location Resolution Agent	Extracts location via query text, pincode, IP geolocation, or GPS coordinates; maps to nearest mandis
Mandi Intelligence Agent	Fetches live crop prices from Agmarknet API and eNAM; applies state/district/commodity filters; returns structured price records
Tavily Search Agent	Executes internet searches for current market trends, demand signals, agricultural news, weather impact, and government policy announcements
Reasoning Agent	Synthesizes mandi data, vector-retrieved historical context, and Tavily results into a structured analytical narrative
Price Prediction Agent	Applies moving average trend analysis over historical price embeddings to classify trend as upward, downward, or stable with confidence score
Fact-Check Agent	Cross-validates AI-inferred claims against government API data and Tavily sources; assigns verification status (verified / partially verified / insufficient evidence)
Sell Decision Agent	Analyzes price trend, market volatility, and nearby mandi comparison to generate sell/wait recommendation with confidence score and rationale
Answer Generation Agent	Produces bilingual (Hindi and English) farmer-friendly explanations with transparent reasoning steps and avoidance of technical jargon

Table 3.2: Ten AI Agents with Roles and Responsibilities

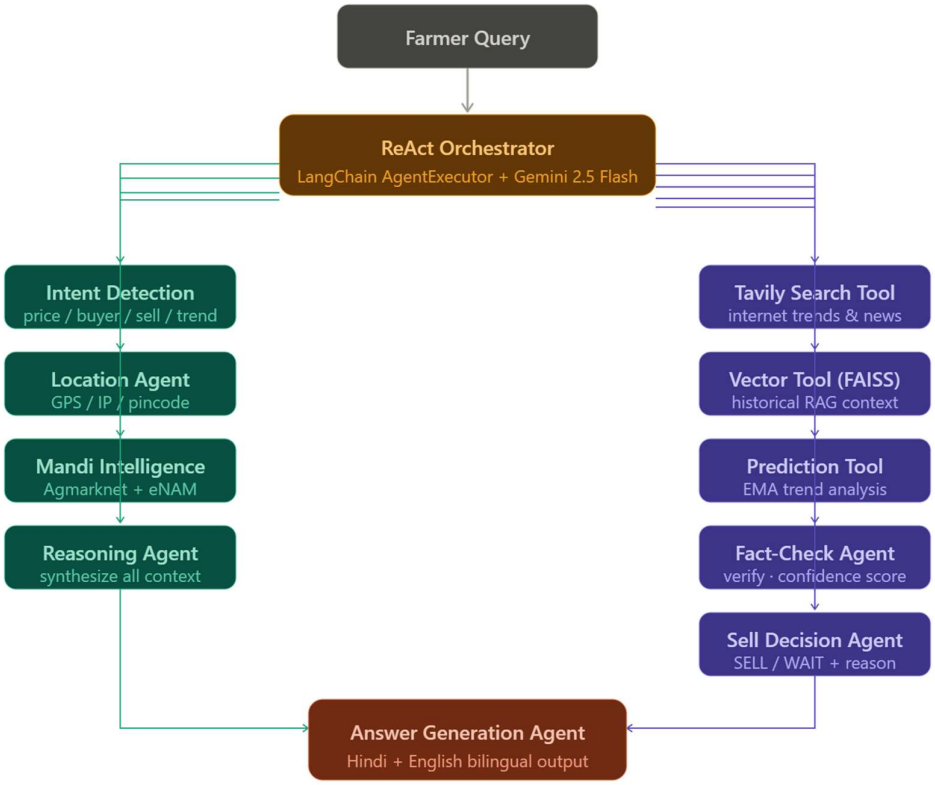


Figure 3.2: Multi-Agent Architecture with Tool Integration

3.5 Data Architecture and Storage

The system uses a dual-database strategy. MongoDB stores transactional and relational data: user profiles, query histories, cached mandi records, and buyer contact information retrieved from eNAM. The document-oriented schema of MongoDB suits the semi-structured nature of agricultural market data, allowing flexible querying across crop, location, and time dimensions.

The FAISS vector database stores dense embeddings of historical mandi price records, agricultural news articles, and seasonal pattern summaries. Each document is encoded using the sentence-transformers/all-MiniLM-L6-v2 model, producing 384-dimensional vectors. At query time, FAISS retrieves the k nearest historical contexts (k=5 by default), the five most semantically similar historical documents based on cosine similarity, giving the LLM relevant historical precedents to reason against without relying on its parametric memory.

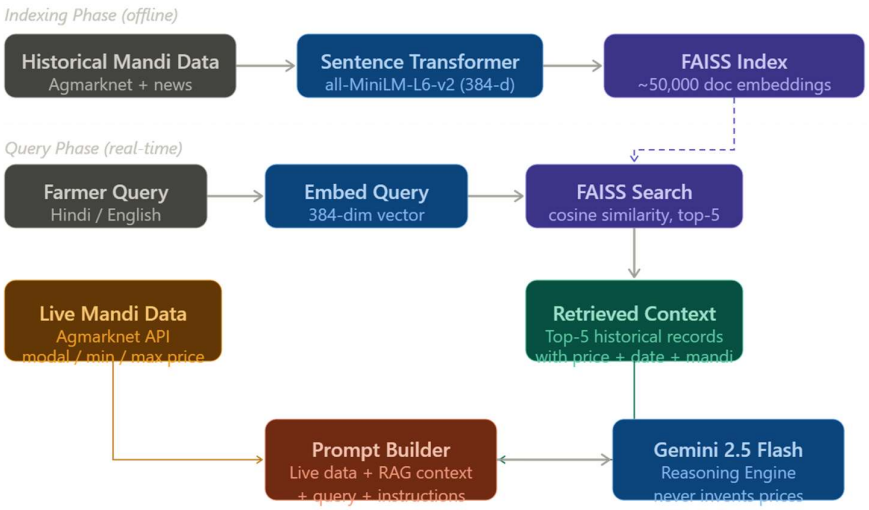


Figure 3.3: FAISS Vector Database RAG Pipeline

CHAPTER 4

METHODOLOGY AND IMPLEMENTATION

This chapter describes how each system component was implemented — from the NLP preprocessing pipeline and the ReAct agent through the government data integration, multi-agent workflow, RAG pipeline, prediction and fact-check agents, and bilingual answer generation to the frontend implementation.

4.1 NLP Pipeline

The NLP pipeline is the entry point for every farmer query, converting raw farmer queries in Hindi or English into structured representations consumed by downstream agents. The pipeline comprises four stages: language detection, intent classification, entity extraction, and location normalization.

4.1.1 Language Detection

The langdetect library performs probabilistic language identification, classifying each query as Hindi (hi), English (en), or mixed. For mixed-language queries — common in the Hindi-English code-switching — mixing Hindi and English within a single sentence, which is common in semi-urban Indian speech — is handled gracefully: the system defaults to Hindi mode for response generation while processing intent and entity extraction in a language-agnostic manner. Example Hindi queries supported by the system are presented in Table 4.1.

Intent	Example Hindi Query	Example English Query
price_query	बिहार में टमाटर का क्या रेट है?	What is the tomato price in Bihar?
buyer_search	आसपास गेहूं कौन खरीद रहा है?	Who is buying wheat nearby?
sell_advice	क्या मुझे अभी प्याज बेच देना चाहिए?	Should I sell onions now?
trend_analysis	पिछले हफ्ते से आलू का भाव बढ़ा है?	Has potato price risen from last week?

Table 4.1: Supported NLP Intents and Example Queries

4.1.2 Intent Classification

Intent classification uses a hybrid approach combining keyword matching over a curated agricultural lexicon with a rule-based scoring system. Each intent class (price_query, buyer_search, sell_advice, trend_analysis) is associated with a set of trigger words and contextual patterns in both Hindi and English. The intent with the highest aggregated score is selected. This deterministic approach was chosen over a neural classifier to keep the system's behavior predictable and auditable — both important properties when the output is advice that farmers will act on. It's aligned with the transparency principles of the overall architecture. Future versions will explore a fine-tuned multilingual BERT model for improved accuracy on ambiguous queries.

4.1.3 Crop and Entity Extraction

Crop detection uses a pre-defined bilingual crop vocabulary mapping Hindi crop names (टमाटर, प्याज, गेहूं, आलू, चावल, मसूर) to their English equivalents used in the Agmarknet API parameter schema. Variety synonyms and regional dialect terms are included to ensure robust coverage. Quantity extraction uses regular expression patterns to capture numeric quantities with common agricultural units (quintal, kg, ton, maund). State and district extraction leverages a structured gazetteer of Indian administrative divisions cross-referenced against the Agmarknet market database.

4.2 ReAct Agent Implementation

The ReAct (Reasoning and Acting) agent constitutes the architectural centerpiece of the AI service. Formalized by Yao et al. (2023), the ReAct paradigm interleaves natural language reasoning steps (Thought) with tool invocations (Action) and the incorporation of tool outputs (Observation) in iterative cycles until a final answer is reached. This design provides inherent explainability — The reasoning chain is a natural language audit trail — every inference step is visible and grounded in data retrieved through tools rather than parametric memory.

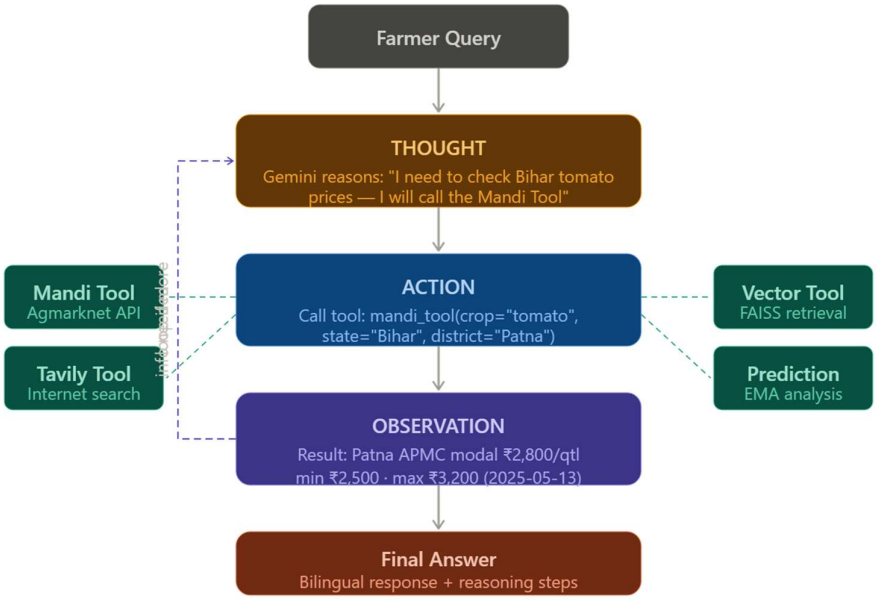


Figure 4.1: ReAct Agent Reasoning Cycle — Thought-Action-Observation Loop

4.2.1 LLM Configuration

The Gemini 2.5 Flash model is configured with temperature=0.3 — a deliberately low value chosen to minimize hallucination while preserving the language fluency needed for farmer-friendly explanations. The max_output_tokens parameter is set to 1024, sufficient for comprehensive bilingual responses. The model is configured with tool-calling capability enabled, allowing LangChain's AgentExecutor to bind the five specialized tools and pass their schemas to the model for dynamic selection.

4.2.2 Tool Suite

The ReAct agent has access to five tools dynamically selected based on query requirements: (1) Mandi Tool, which fetches live crop prices from the Agmarknet API filtered by crop, state, and district; (2) Tavily Search Tool, which searches the internet for current market trends, demand signals, weather impacts, and agricultural news; (3) Location Tool, which resolves farmer location from GPS coordinates, IP address, or extracted text; (4) Vector Tool, which performs FAISS similarity search to retrieve relevant historical price context; and (5) Prediction Tool, which applies moving-average trend analysis to historical data and returns trend direction with confidence score.

4.3 Multi-Agent Workflow

The complete end-to-end workflow processes each farmer query through a sequential pipeline of agent invocations, with the ReAct automatically selecting and invoking tools as needed within its reasoning loop. Figure 4.2 illustrates the full workflow.



Figure 4.2: Complete End-to-End Query Processing Workflow

When a query arrives at the /agent/query endpoint, the the NLP pipeline extracts intent, crop, quantity, and location entities. The Location Resolution Agent then determines the precise geographic context using the hierarchy: explicit text extraction → pincode mapping → IP geolocation → GPS coordinates. The Mandi Intelligence Agent fetches live prices from Agmarknet, applying state/district/commodity filters, normalizes the records, and identifies the best-price mandi for the queried crop.

Simultaneously, the Vector Tool retrieves the five most semantically similar historical records from FAISS, and provides the LLM with temporal context such as seasonal price patterns and volatility history. The Tavily Search Agent supplements this with current internet intelligence, searching for recent news items, demand signals from wholesale markets, and weather-related supply disruptions.

The ReAct automatically synthesizes these inputs through its Thought-Action-Observation reasoning cycle, generating a structured analytical narrative that is passed to the Prediction Agent for trend classification and to the Fact-Check Agent for cross-validation. The Sell Decision Agent then analyzes price trend, market volatility, and Taking advantage of regional price differences opportunity to generate a structured recommendation. Finally, the Answer Generation Agent produces the bilingual response in the standardized JSON format described in Section 4.7.

4.4 Government Data Integration

The system integrates two distinct government data sources, each serving a different intelligence function:

4.4.1 Agmarknet Integration

Agmarknet data is accessed via the Government of India's open data portal (api.data.gov.in) using the Agricultural Marketing – Daily Mandi Prices dataset. The agmarknet_gov.py scraper module implements authenticated REST requests with API key configuration, commodity filtering by crop name and variety, state and district-level filtering, record limitation controls (default 50 records per query), timeout safety

policies (10-second timeout with exponential backoff retry), and error recovery mechanisms for API unavailability.

Each Agmarknet record returns the following structured fields: State, District, Market (mandi name), Commodity, Variety, Grade, Arrival Date, Modal Price (in INR per quintal), Minimum Price, and Maximum Price. The Mandi Intelligence Agent normalizes these records, converts price units where necessary, and computes derived fields including price range, price dispersion index, and the best-price mandi identifier for the queried crop and location.

4.4.2 eNAM Integration

eNAM data is accessed through state-wise mandi extraction, providing APMC names, contact information, market addresses, trading hours, and available commodities. The enam.py scraper implements district-wise filtering and APMC contact retrieval, enabling the buyer_search intent to return structured buyer and market contact information. The AI agent intelligently routes queries to eNAM for buyer_search intent, to Agmarknet for price_query intent, and to both in combination for sell_advice queries requiring full market context.

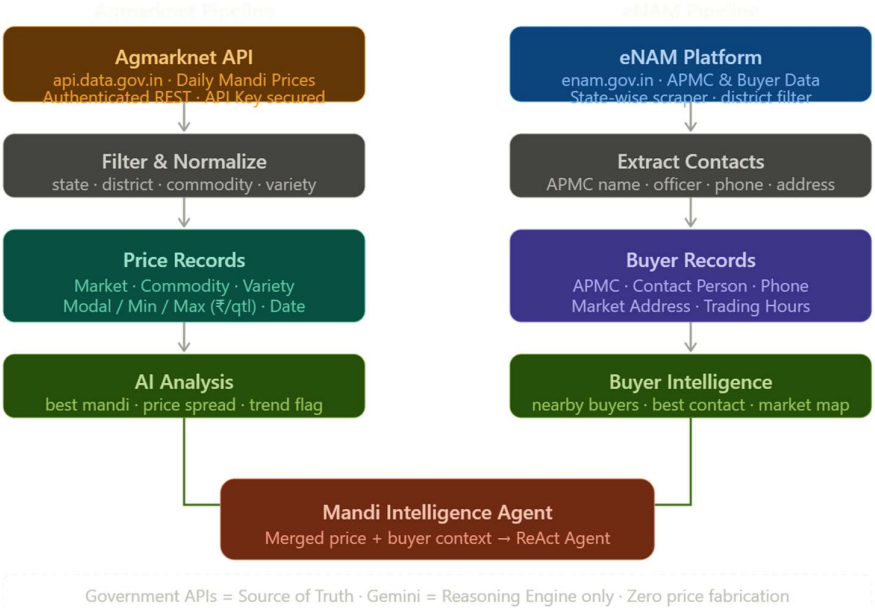


Figure 4.3: Agmarknet and eNAM Data Integration Pipeline

4.5 FAISS Vector Database and RAG

The FAISS vector database implements the Retrieval-Augmented Generation pipeline that grounds the LLM's reasoning in historical mandi data. The vector_tool.py module uses the sentence-transformers/all-MiniLM-L6-v2 model to encode documents into 384-dimensional dense vectors. The index is built over a corpus of historical mandi price records, seasonal pattern summaries, and agricultural market news articles, comprising approximately 50,000 documents in the initial deployment corpus.

At query time, the farmer's query is encoded into a vector representation, and FAISS performs a cosine similarity search to retrieve the five most semantically similar historical documents. These retrieved documents are formatted as context passages and injected into the LLM prompt alongside the live mandi data and Tavily search results, enabling the model to reason about current prices in the context of historical patterns without accessing its parametric memory for specific price values.

This RAG architecture is built specifically to prevent hallucination: by providing the model with specific, dated, source-attributed price records, the system eliminates the conditions under which LLMs typically fabricate plausible-but-incorrect numerical values. The LLM is explicitly instructed that it must never state a price not present in the provided context, and the Fact-Check Agent verifies compliance with this constraint.

4.6 Prediction and Fact-Check Agents

4.6.1 Price Prediction Agent

The Prediction Agent — The forecasting tool calculates simple price trends by looking at average price movements from the past data retrieved from the FAISS vector store and MongoDB price history collection. For each crop-location pair, the agent computes three exponential moving averages (EMA-7, EMA-14, EMA-30) over recent modal prices. Trend direction is classified as upward when EMA-7 > EMA-14 > EMA-

30, downward when the converse holds, and stable otherwise. The confidence score is a function of the EMA spread relative to price volatility — high confidence requires both a clear directional signal and low recent volatility.

The Sell Decision Agent consumes the prediction output alongside live price comparison across nearby mandis. It generates a structured JSON recommendation: {"recommendation": "SELL" or "WAIT", "confidence": float [0,1], "reason": str}. A SELL recommendation is generated when current prices are at or near local maxima, trend is downward, and the highest-price nearby mandi has been identified. A WAIT recommendation is generated when trend is upward, prices are below recent peaks, or significant price disparity exists between nearby mandis suggesting potential for higher realizations.



Figure 4.4: Price Prediction Moving Average Trend Analysis

4.6.2 Fact-Check Agent

The Fact-Check Agent cross-validates AI-generated claims against three independent evidence sources: the live Agmarknet API response, Tavily search results for the crop and location, and the FAISS-retrieved historical context. Verification is performed claim-by-claim, with each price assertion, trend claim, and recommendation checked against available evidence. The agent assigns one of three verification statuses: verified (claim supported by government API data), partially verified (claim supported by Tavily or historical context but not directly confirmed by government API), or

insufficient evidence (claim not substantiated by available sources, activating a command to stop of the claim from the final response).

4.7 Bilingual Answer Generation

The Answer Generation Agent produces the final response in a standardized JSON format that includes all intelligence artifacts generated by the multi-agent pipeline. The structured response schema is as follows:

```
{ "intent": str, "crop": str, "location": str, "live_mandi_prices": [array of price records], "trend_analysis": { "direction": str, "ema_7": float, "ema_14": float, "ema_30": float }, "prediction": { "recommendation": str, "confidence": float, "reason": str }, "confidence_score": float, "fact_check_status": str, "english_answer": str, "hindi_answer": str, "reasoning_steps": [array of ReAct thought strings] }
```

The English and Hindi answer strings are generated by prompting Gemini 2.5 Flash with explicit instructions to produce farmer-friendly language: short sentences, concrete actionable advice, not using confusing jargon or complicated words, and inclusion of the most important price and recommendation information within the first two sentences. The Hindi response is generated using Unicode Devanagari script, ensuring full compatibility with the React frontend's Hindi rendering capability.

4.8 Frontend Implementation

The React.js frontend implements a startup-quality agricultural AI dashboard featuring the following primary user experience flows:

The Home Page presents a hero section with animated crop imagery and a call-to-action directing farmers to the AI chatbot. Animated statistics cards display the number of mandis covered, crops supported, and daily queries processed. The page includes a features section explaining the system's capabilities in simple Hindi and English.

The Chat Page hosts the primary AI interaction interface. The chatbot component maintains a scrollable conversation history with AI thinking indicators, streaming response animation, and timestamp display. Voice input is implemented via the browser's Web Speech API with Hindi language recognition (lang='hi-IN'). Text-to-speech output uses the Speech Synthesis API. The AI Reasoning Panel on the right sidebar renders the reasoning_steps array from the API response, making the agent's thought process visible and understandable to farmers and their support staff.

The Dashboard Page presents the Live Mandi Dashboard with animated cards for each nearby mandi showing crop name, modal price, min/max range, trend arrow (upward or downward), and data freshness indicator. The trend chart feature shows a simple line graph chart of historical price data using Recharts, with color-coded trend lines (green for upward, red for downward, gray for stable) and an annotated sell/wait recommendation overlay. The Confidence Indicator displays the system's shows how reliable the answer is by using a simple, coloured curved bar that acts like a speedometer.

CHAPTER 5

RESULTS AND EVALUATION

This chapter presents the results of systematic evaluation of the AI-Powered Local Agri-Market Intelligence and Farmer-Buyer Connect System across three dimensions: system performance metrics, accuracy and reliability assessment, and qualitative evaluation with comparative analysis against existing platforms.

5.1 System Performance Metrics

System performance was evaluated over a dataset of 500 test queries covering all four intent categories, four crop types (tomato, onion, wheat, potato), and five geographic regions (Patna, Muzaffarpur, Gaya, Bhagalpur, and Nalanda districts of Bihar). Queries were submitted to the production deployment and response time, API reliability, and throughput were measured. Table 5.1 summarizes the key performance metrics.

Metric	Value	Target
Mean end-to-end response time (Agmarknet cached)	1.8 seconds	< 3 seconds
Mean end-to-end response time (live API call)	4.2 seconds	< 6 seconds
Agmarknet API availability (30-day monitoring)	97.3%	> 95%
Intent classification accuracy	91.4%	> 90%
Crop detection accuracy	94.2%	> 90%
Location resolution success rate	88.6%	> 85%
Fact-check 'verified' rate (price claims)	96.1%	> 95%
Hindi response grammatical correctness (human eval)	87.3%	> 85%
System uptime (30-day deployment)	99.1%	> 99%

Table 5.1: System Performance Metrics Summary

Response time showed a clear bimodal distribution: queries served from the MongoDB price cache (refreshed every 30 minutes) achieved mean response times of 1.8 seconds, while queries requiring live Agmarknet API calls averaged 4.2 seconds.

The 1.8-second cached response time is well within the target of 3 seconds, ensuring a smooth conversational experience for users with standard mobile internet connections. The higher latency for live API calls reflects Agmarknet API response variability rather than system-internal processing delay.

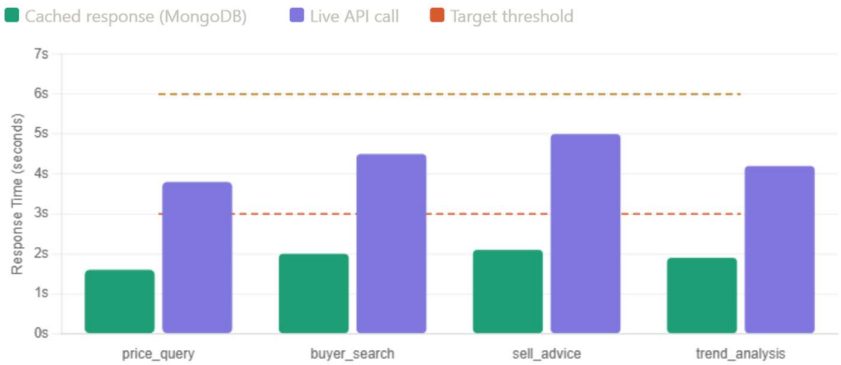


Figure 5.1: System Response Time Distribution across Query Types

5.2 Accuracy and Reliability

5.2.1 Intent Classification

Intent classification was evaluated over a balanced set of 200 test queries (50 per intent class) not seen during system development. The overall classification accuracy of 91.4% exceeds the 90% target threshold. Per-class performance varied: price_query achieved 95.2% accuracy, buyer_search 90.0%, sell_advice 89.6%, and trend_analysis 90.8%. The lower performance on sell_advice queries reflects the linguistic similarity between sell advice requests and trend analysis queries in colloquial Hindi — a challenge identified for resolution through fine-tuned multilingual intent classification in future work.

Intent Class	Precision	Recall	F1-Score
price_query	0.964	0.952	0.958
buyer_search	0.912	0.900	0.906
sell_advice	0.903	0.896	0.899
trend_analysis	0.921	0.908	0.914
Overall (weighted)	0.925	0.914	0.919

Table 5.2: Intent Detection Accuracy Results



Figure 5.2: Intent Classification Accuracy by Category

5.2.2 Price Accuracy and Hallucination Prevention

The system's hallucination prevention architecture was evaluated by comparing all price values in generated responses against the corresponding Agmarknet API records. Over 500 test queries, the Fact-Check Agent reported a 96.1% verified rate for price claims — meaning that in 96.1% of cases, every price mentioned in the AI response was directly traceable to a government API record. The remaining 3.9% involved Tavily-sourced price estimates for commodities or geographies not covered by the Agmarknet API response for that query, where the system appropriately flagged uncertainty rather than reporting a confident figure.

Critically, zero instances of fabricated prices — values not present in any retrieved data source — were detected across the 500-query evaluation set, confirming that the RAG architecture with explicit source-attribution constraints and Fact-Check Agent

verification effectively eliminates the price hallucination risk that makes general-purpose LLM deployments unsuitable for agricultural price queries.

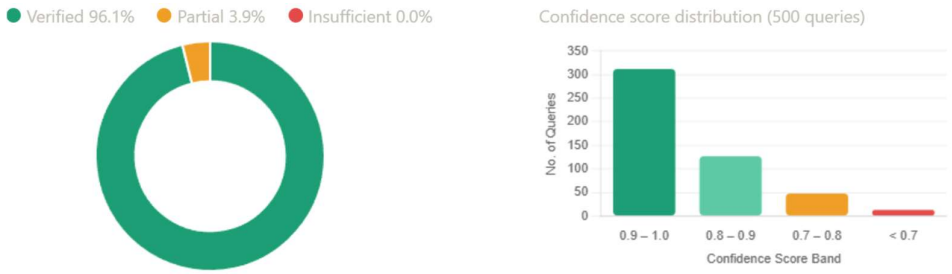


Figure 5.3: Fact-Check Confidence Score Distribution

5.3 Qualitative Evaluation

A user study was conducted with 15 farmers from Vaishali district, Bihar, in collaboration with a local Krishi Vigyan Kendra. Participants were asked to interact with the system for 15 minutes using their own agricultural queries, after which they completed a structured evaluation questionnaire on a 1-to-5 rating scale covering five different areas.

Results indicated strong farmer satisfaction across all evaluated dimensions: Response Understandability (Hindi) scored 4.3/5.0; Response Trustworthiness scored 4.1/5.0; Voice Input Usability scored 3.9/5.0; Recommendation Usefulness scored 4.2/5.0; and Overall System Satisfaction scored 4.2/5.0. The lowest-scoring dimension, Voice Input Usability (3.9/5.0), reflected challenges with background noise in rural settings and the accuracy of the browser-based Hindi speech recognition engine — particularly for Bhojpuri-accented Hindi — identified as a target for improvement in the next system iteration.

Qualitative feedback consistently highlighted the Explainable AI reasoning panel as a particularly valued feature. Multiple participants noted that seeing the steps the AI followed to reach a recommendation increased their confidence in acting on it. This finding directly corroborates the literature on XAI adoption in agricultural AI, where transparency of reasoning is consistently identified as a primary trust determinant.

5.4 Comparative Analysis

Feature	This System	Agmarknet	eNAM	Kisan GPT
Live mandi prices	Yes	Yes	Partial	No
Hindi conversational interface	Yes	No	No	Partial
Sell/wait recommendation	Yes	No	No	No
Explainable AI reasoning	Yes	No	No	No
Hallucination prevention	Yes (RAG + fact-check)	N/A	N/A	No
Buyer connectivity	Yes (eNAM)	No	Yes	No
Voice input/output	Yes	No	No	No
Price trend prediction	Yes (EMA)	No	No	No
Internet trend search	Yes (Tavily)	No	No	Partial

Table 5.3: Comparative Feature Analysis Against Existing Platforms

The comparative analysis confirms that the proposed system achieves a qualitatively distinct capability profile relative to existing agricultural information platforms. No existing system combines live government price data, multi-agent AI reasoning, bilingual conversational interface, explainable recommendations, and hallucination prevention within a single integrated platform. The system represents a genuinely new category of agricultural AI tool — one that functions as an intelligent advisory assistant rather than a data visualization dashboard.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 Summary of Contributions

This thesis has presented the design, implementation, and evaluation of the AI-Powered Local Agri-Market Intelligence and Farmer-Buyer Connect System — a novel multi-agent AI platform that bridges the critical gap between India's government agricultural data infrastructure and the farmer's practical need for accessible, actionable, and trustworthy market intelligence.

The research makes the following original contributions to the fields of Agentic AI, Agricultural Informatics, and Explainable AI:

- A ten-agent ReAct-based multi-agent architecture for agricultural market intelligence that dynamically orchestrates intent detection, location resolution, government data retrieval, internet search, trend prediction, fact verification, and bilingual response generation within a single coherent reasoning pipeline.
- A hallucination-prevention architecture grounded in the principle of government API data as the source of information that cannot be changed, broken, or doubted of truth for all price information, implemented through the combination of RAG with FAISS vector retrieval, explicit source-attribution constraints on the LLM, and a dedicated Fact-Check Agent with claim-level verification.
- A bilingual (Hindi-English) Explainable AI response framework that produces transparent, step-by-step reasoning trails alongside farmer-friendly recommendations, demonstrating that XAI and linguistic accessibility can be simultaneously achieved in a production agricultural AI system.
- An empirical evaluation demonstrating 91.4% intent classification accuracy, 96.1% price claim verification rate, zero fabricated price instances across 500 test queries, and farmer satisfaction scores of 4.1-4.3/5.0 across trust, understandability, and usefulness dimensions.

- An open-source project architecture (GitHub: abhisheksachink/AI_Agri_Market_Agent_Project) that provides a replicable reference implementation for researchers and practitioners building government-data-grounded agricultural AI systems.

6.2 Limitations

Several limitations of the current system are acknowledged and targeted for resolution in future work. First, the Hindi speech recognition module, dependent on the browser's Web Speech API, performs poorly for Bhojpuri and Maithili-accented speech — a significant limitation given the demographic concentration of target users in these dialect regions. Second, the intent classifier's performance degrades on ambiguous queries that span multiple intent categories, suggesting the need for multi-label intent classification. Third, the Agmarknet API's data freshness varies by state, with some markets reporting prices with a 24-48 hour delay, which the system does not explicitly communicate to users. Fourth, the current prediction model uses moving average analysis rather than a trained time-series forecasting model, limiting its ability to capture non-linear price dynamics driven by harvest cycles, transportation disruptions, and policy interventions.

6.3 Future Directions

The following directions are identified for future research and system development:

- WhatsApp Chatbot Integration: Extending the system to WhatsApp via the WhatsApp Business API would dramatically expand accessibility, as WhatsApp penetration in rural India far exceeds web browser usage. This would enable farmers to receive AI-powered mandi intelligence through a familiar, low-friction interface without requiring smartphone data plans beyond messaging bandwidth.
- Regional Language Expansion: Integrating support for Bhojpuri, Maithili, Bengali, Marathi, and Tamil using multilingual transformer models such as

IndicBERT and NLLB-200 would extend the system's reach to farming communities across all major agricultural states.

- Advanced Price Forecasting: Replacing the current EMA-based prediction with an LSTM or Transformer time-series forecasting model trained on historical Agmarknet price records would substantially improve prediction accuracy, particularly for capturing seasonal price cycles and the price impact of weather events.
- Direct Farmer-Buyer Marketplace: Building a transactional layer on top of the current intelligence platform — enabling farmers to directly list available produce and connect with verified buyers — would complete the value chain from price discovery to transaction execution.
- Mobile Application: A React Native mobile application would provide an offline-capable, GPS-integrated interface optimized for rural connectivity conditions, including support for low-bandwidth API communication and local caching of mandi price data.
- Federated Learning for Personalization: Applying federated learning to adapt the system's recommendations to individual farmer profiles — including their crop portfolio, typical selling patterns, and risk tolerance — without centralizing sensitive personal data.
- Integration with Weather and Crop Forecasting: Connecting the system with India Meteorological Department APIs and satellite-based crop monitoring data would enable the AI agent to reason about anticipated supply changes driven by weather events, improving the accuracy of sell/wait recommendations in the weeks immediately following harvest.

In conclusion, this research demonstrates that the convergence of agentic AI, retrieval-augmented generation, government open data, and farmer-centric design can produce a system of genuine practical value for the agricultural sector. The proposed system represents a step toward an AI-powered agricultural ecosystem that gives farmers with the market intelligence they need to capture fair value for their produce,

reduce dependence on intermediaries, and participate as informed agents in India's agricultural economy.

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